HPE Labs

HPE Cray EX4000 Platform & NVIDIA 4-way GH200 Superchip: Hardware and Software

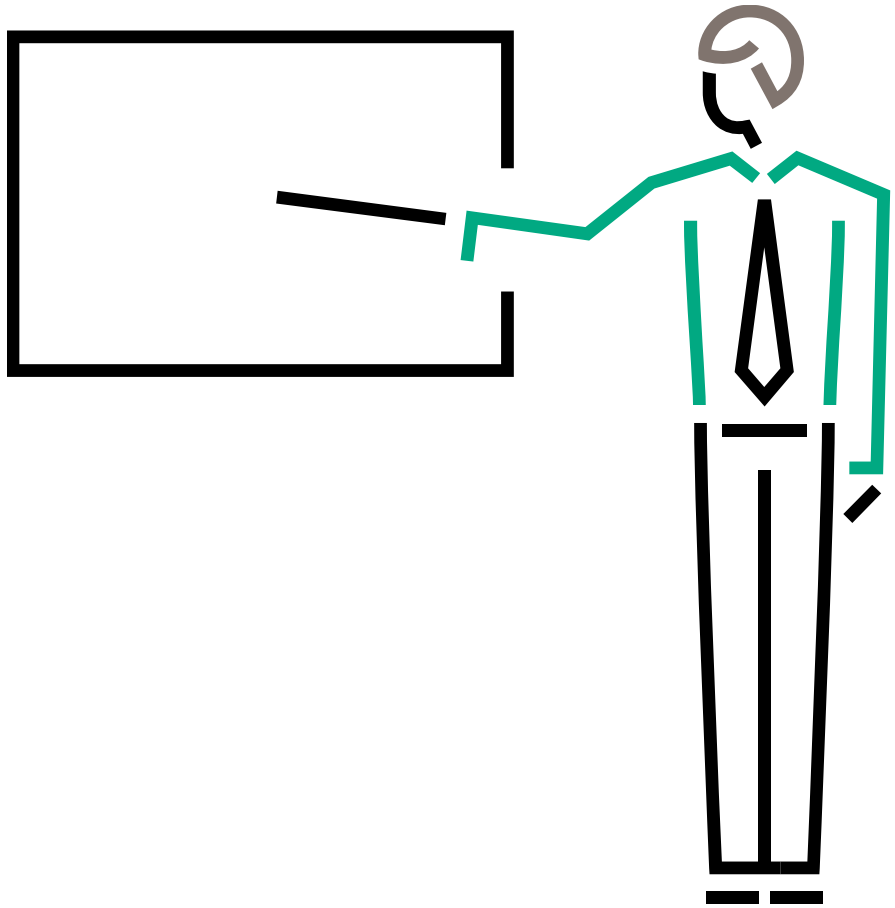
Tim Dykes, EMEA Research Lab, HPE

2025-09-02

HPE

Agenda

- Isambard AI Recap
- HPE Cray EX4000 Architecture
- GH200 Architecture
- HPE Software Stack for EX4000
- NVIDIA Software Stack for GH200



5MW HPE POD is the new national AI supercomputer of the United Kingdom

HPE provided the total carbon emissions of the whole solution at scope 3, from IT to facilities, on a five-year operations timeline

- Warm water, exiting at almost 50°C/122°F, will be used on a highly energy efficient heat re-use model
- The system is designed for waste heat to be extracted and reused to be re-used as renewable energy to heat local buildings
- Supports net zero carbon efficiency targets for 2030
- PUE is 1.1

Currently

#4

on the Green500 list of supercomputers*

HPE HPC/AI infrastructure

- 5,280 GH200 NVIDIA superchips in HPE Cray Supercomputing EX
- 21 ExaFLOP/s of AI performance
- 22 PB Lustre-based all flash file system



"It was **48 hours** from an empty concrete pad to a data center."

Professor Simon Macintosh-Smith, Director of the Bristol Centre for Supercomputing (BriCS), University of Bristol

*As of Greeen500 list June 2025

HPE Cray EX4000

HPE Cray EX : 6 of the world's Top 10 supercomputers*

#1



Lawrence Livermore National Laboratory

#2



OAK RIDGE National Laboratory

#3



Argonne NATIONAL LABORATORY

#6



eni

#8



CSCS

#9



EuroHPC Joint Undertaking

*Top500.org, Top500 June 2025

Recipe for a Supercomputer

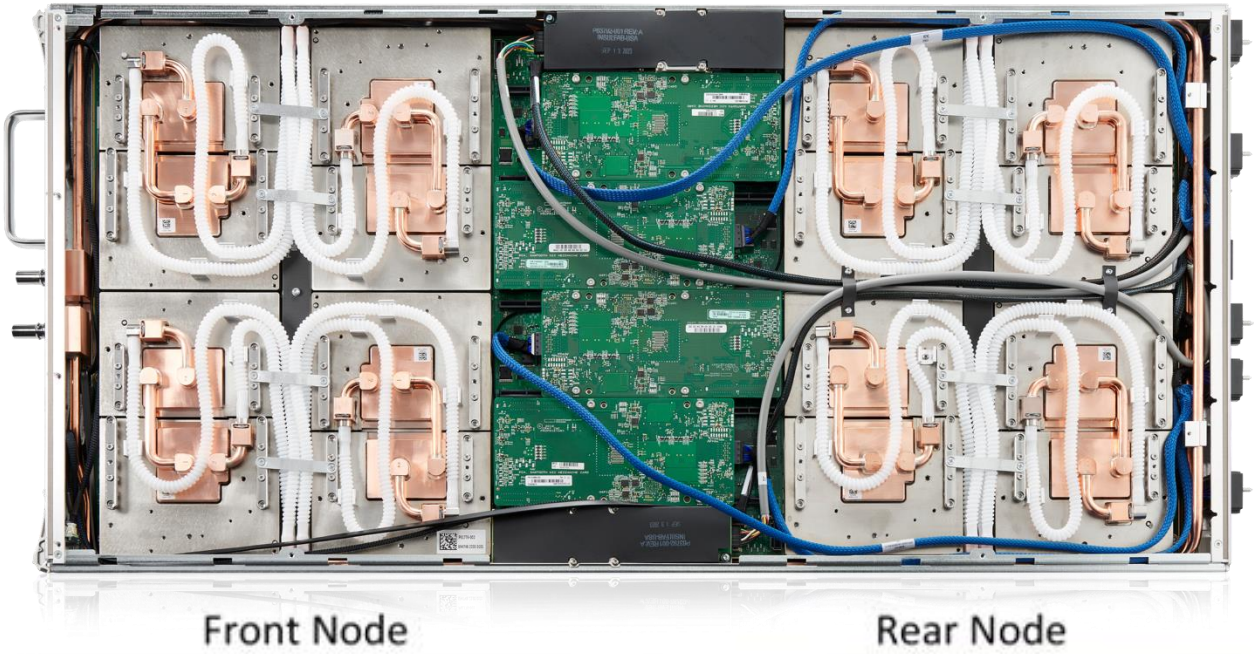
1. Select appropriate microprocessor
   - Function of time and use case
2. Surround it with a bandwidth-rich environment
   - Local memory
   - Interconnection network
3. Scale the system
   - Provide scalable programming and performance tools
   - Provide scalable I/O
   - Eliminate operating system interference (OS jitter)
   - Design in reliability and resiliency
   - Provide scalable system management
   - System service life

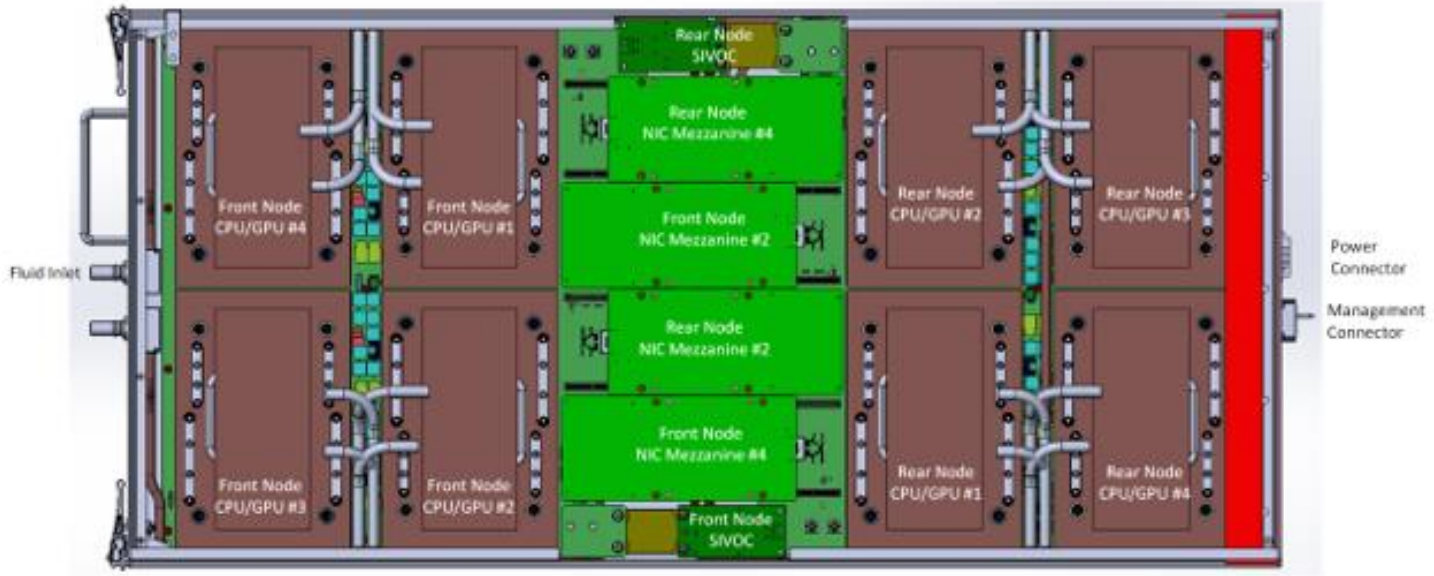
> "Anyone can build a fast CPU. The trick is to build a fast system."
>
> — *Seymour Cray*



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1. Compute Blade Architecture: EX254n Accelerator Blade





- Two 4-socket Nvidia GH200 Grace Hopper Superchip nodes per blade
- 96GB HBM3 per GPU; 120GB LPDDR per CPU
- Up to 8 HPE Slingshot 200Gbit/sec ports per blade
- 0 or 1 local NVMe M.2 SSD per node (up to 2 per blade)
- 2 Board Management Controllers (BMC) per blade
- Cooled with cold plate

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2. Connecting nodes together: Slingshot

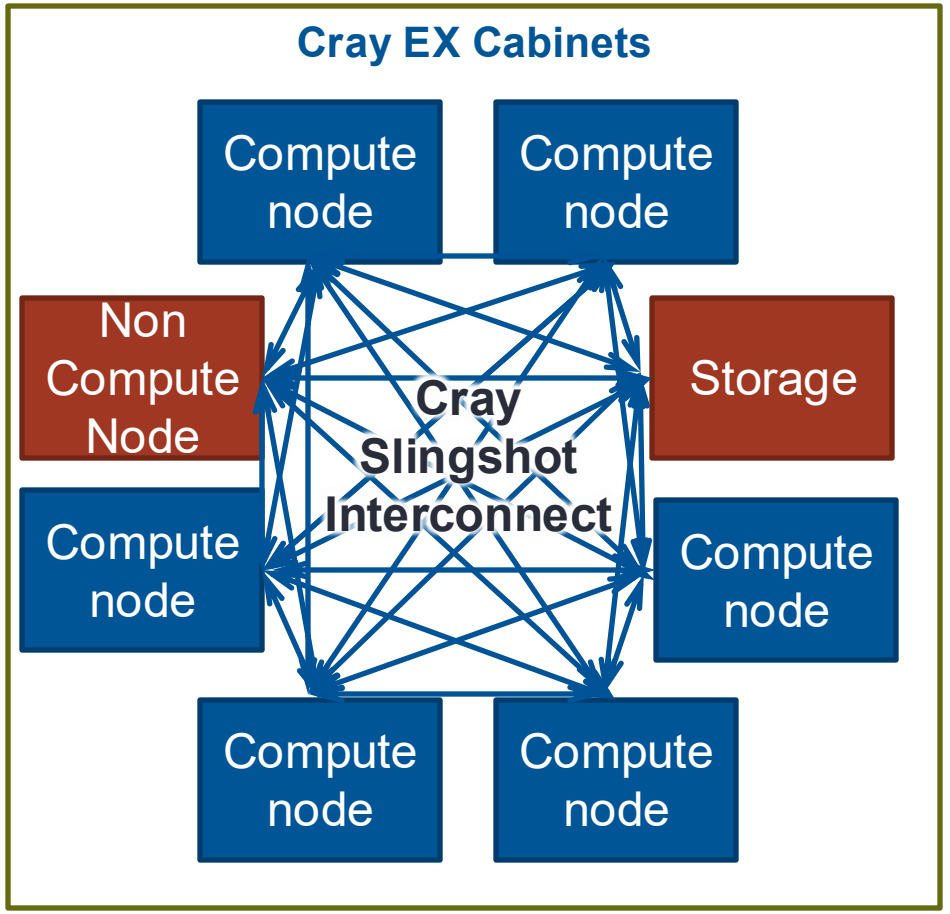
To function as a single efficient supercomputer, individual nodes must be efficiently networked.

- Nodes are interconnected by the high-speed low-latency Slingshot network

HPE Slingshot focus – The network interconnect for the Exascale era

- Industry leading performance and scalability
- Hi-radix, 64 port x 100GbE / 200GbE interfaces
- 12.8 Tb/s bandwidth switch
- Scalable to 250,000+ host ports
- Innovative hardware congestion management, adaptive routing, and quality of service
- Ethernet standards and protocols, optimized HPC functionality
- Standardized, open API management interfaces



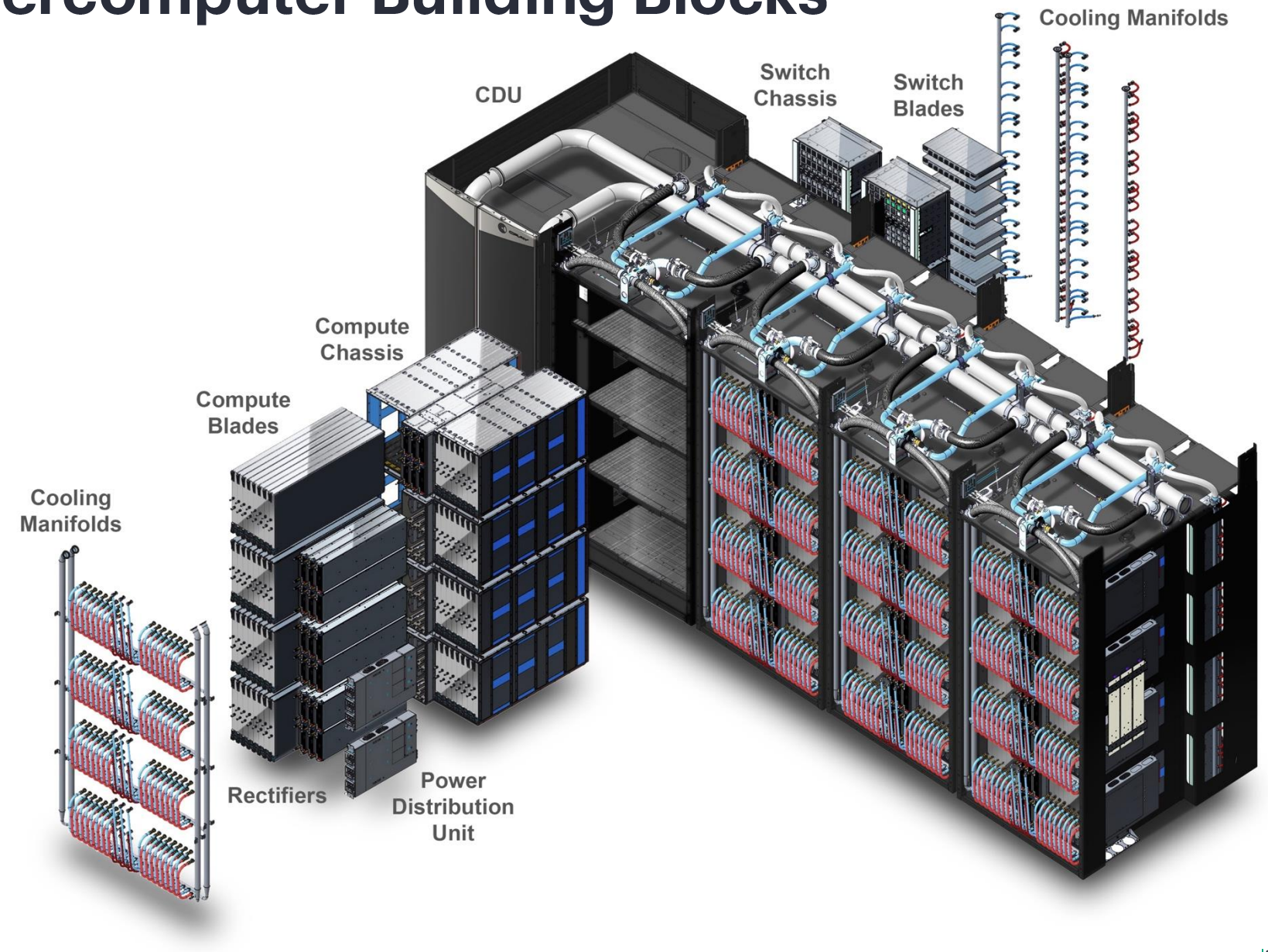


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3. HPE Cray EX Supercomputer Building Blocks



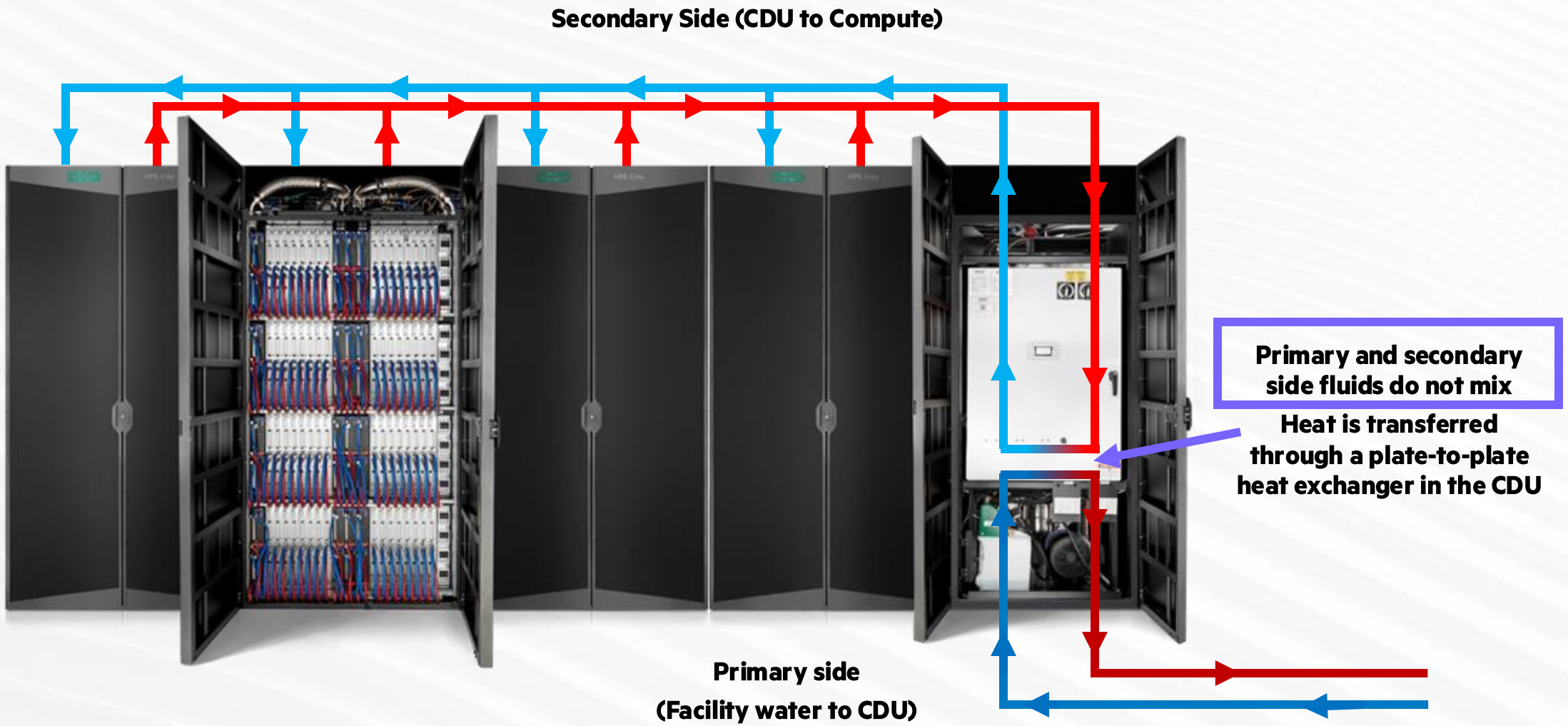
Cooling Distribution Unit



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100% fanless direct liquid cooled HPE Cray Supercomputing EX4000

Unprecedented performance density: Up to 98,304 CPU cores or up to 448 GPUs per cabinet

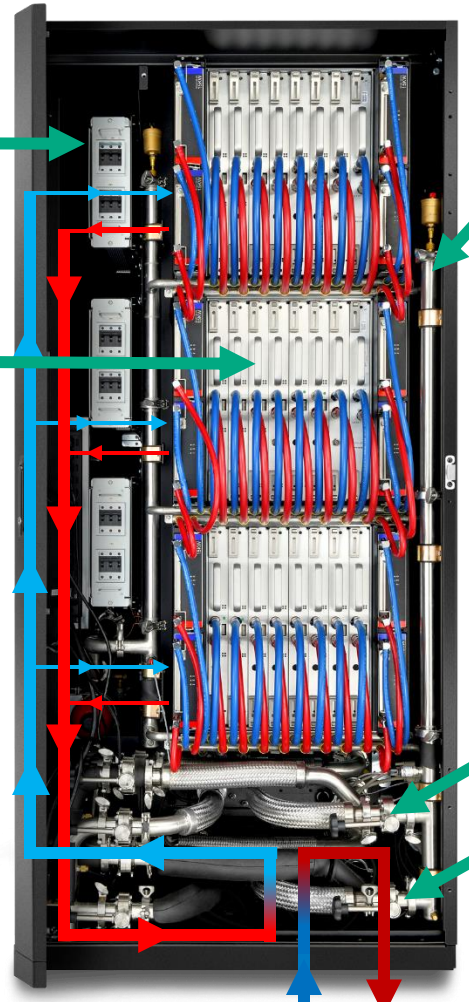


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Isambard AI Phase 1: HPE Cray Supercomputing EX2500

Up to 168 NVIDIA GH200 Superchips in a Single Rack

Front View



PDUs

Blade and Switch Chassis

Secondary Side (CDU to Compute)

Water Manifolds

Primary and Secondary Side fluids do not mix.

Heat is transferred through a plate-to-plate heat exchanger in the CDU

1 CDU for one Chassis or 2 CDUs for 2 or more Chassis

Primary Side (Facility Water to CDU)



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Cableless Network Connection to Liquid Cooled HPE Slingshot Fabric Switches

QSFP-DD Cables

Group cables - PEC

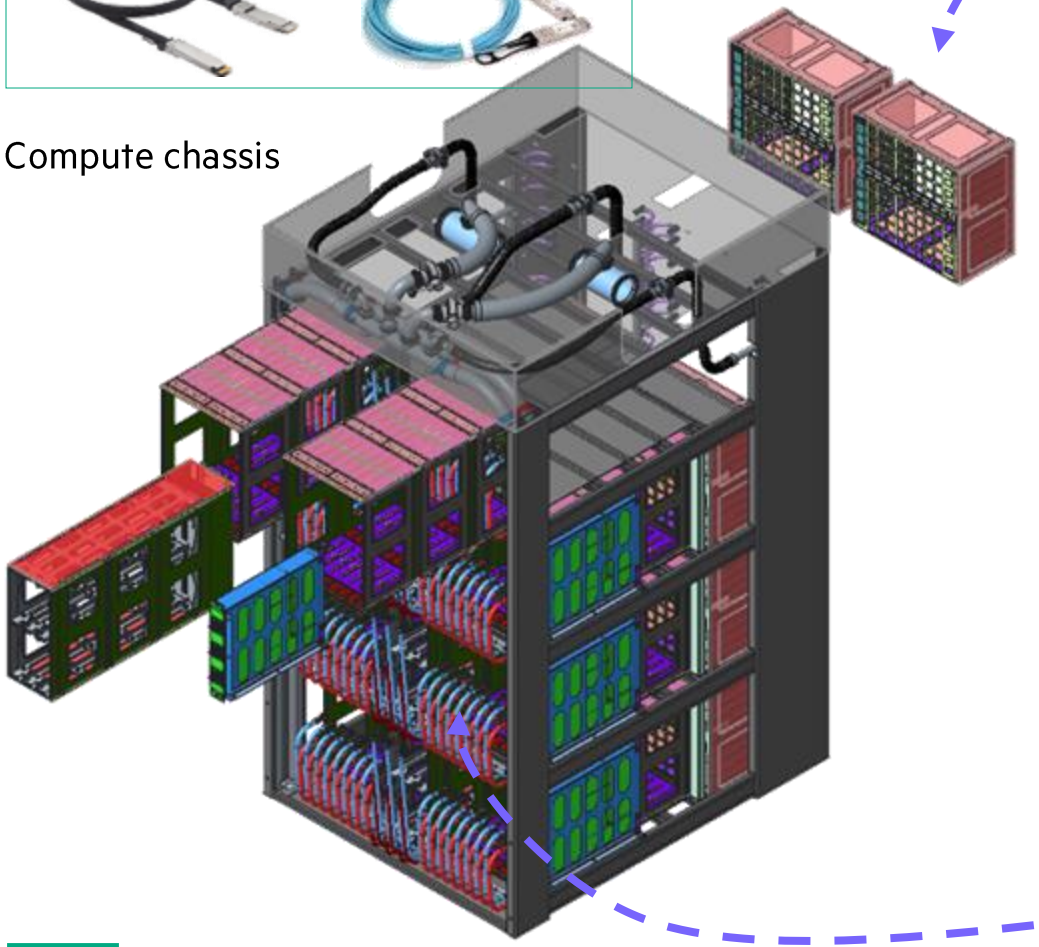


Global cables - AOC



Switch Chassis

Compute chassis



HPE Slingshot Switch (Direct Liquid Cooled)

- 64 ports @ 200 Gbps Per Port (in 32 DD jacks)
- Up to four (4) Slingshots switches per chassis (32 switches per cabinet)

Rear - to fabric



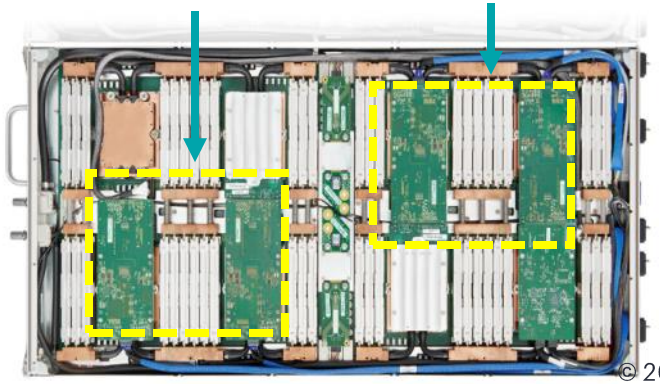
Group/Global links

Internal – mate to blade



NIC Mezzanine Slingshot 11 (200 Gbps)

- HPE Cray Supercomputing EX425 / EX4252 Compute Blades (1 or 2 Injection ports per node)
- HPE Cray Supercomputing EX255a / EX254n GPU Accelerated Blades (4 injection ports per node)



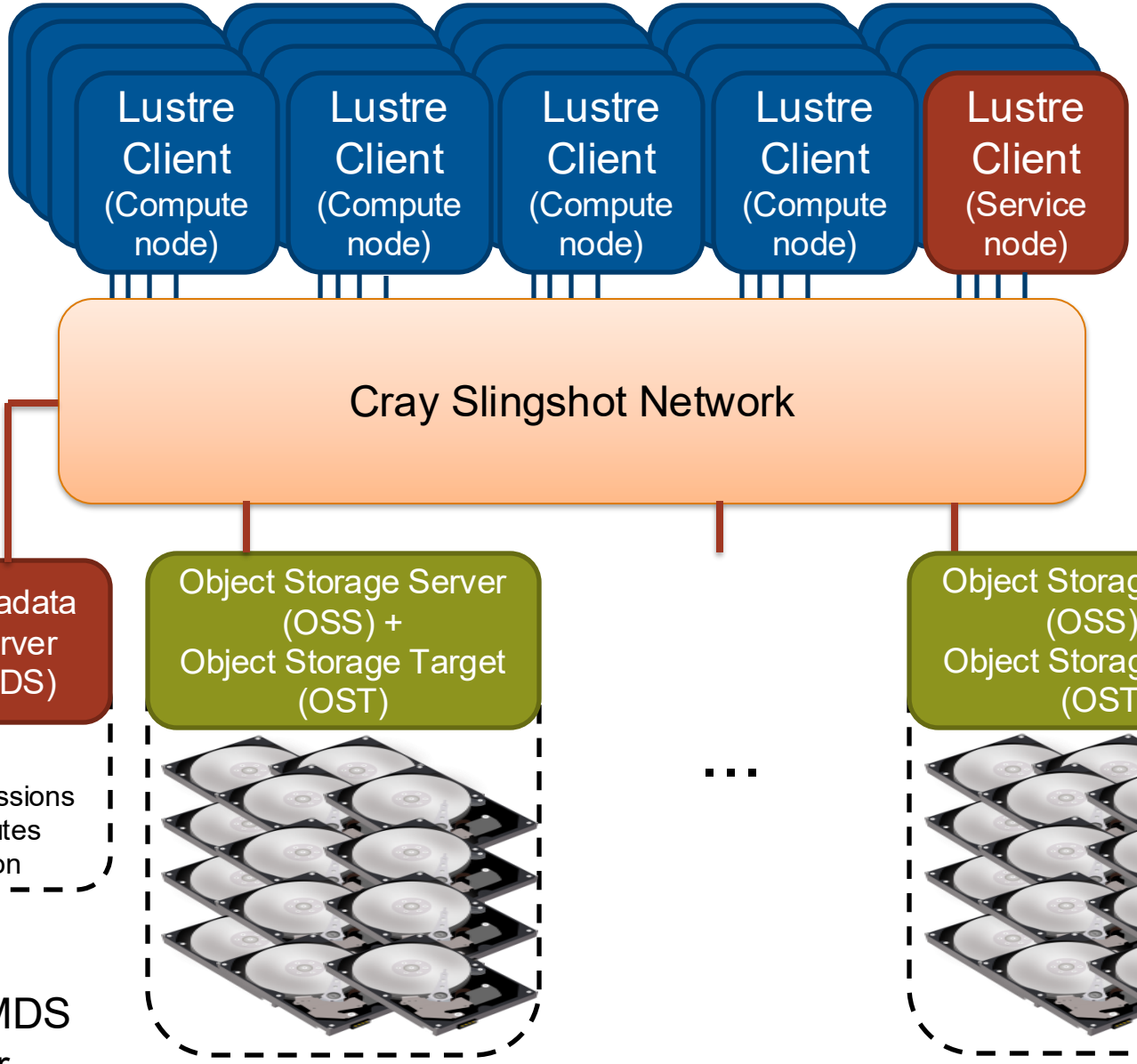
HPE Cray EX4252 Compute Blade

Switch and NICs mate mid-cabinet at 90˚ orientation

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Storage





One MDS per filesystem

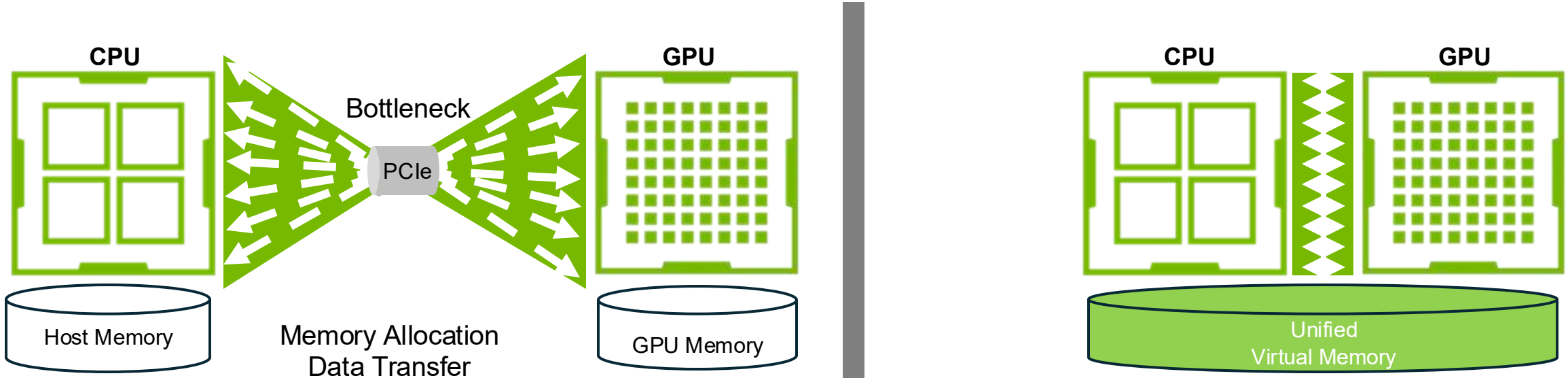


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NVIDIA 4-Way GH200 Superchip

Challenges with Traditional Accelerated Systems

Bridging the GAP: Unifying CPU and GPU Memory



PCIe bottlenecks CPU-GPU communication
GPU can't access CPU memory directly
Complex Programmability
Low CPU Performance per Watt

Legacy PCIe Architecture

We need a new chip architecture:

Accelerate CPU-GPU data transfers
Coherent CPU and GPU cache memory
Simplify Programmability
Deliver more performance per Watt

NVIDIA

NVIDIA Grace CPU Superchip

2X Performance at the Same Power for the Modern Data Center

High Performance Power Efficient Cores

144 flagship Arm Neoverse V2 Cores with
SVE2 4x128b SIMD per core

Fast On-Chip Fabric

3.2 TB/s of bi-section bandwidth connects
CPU cores, NVLink-C2C, memory, and system IO

High-Bandwidth Low-Power Memory

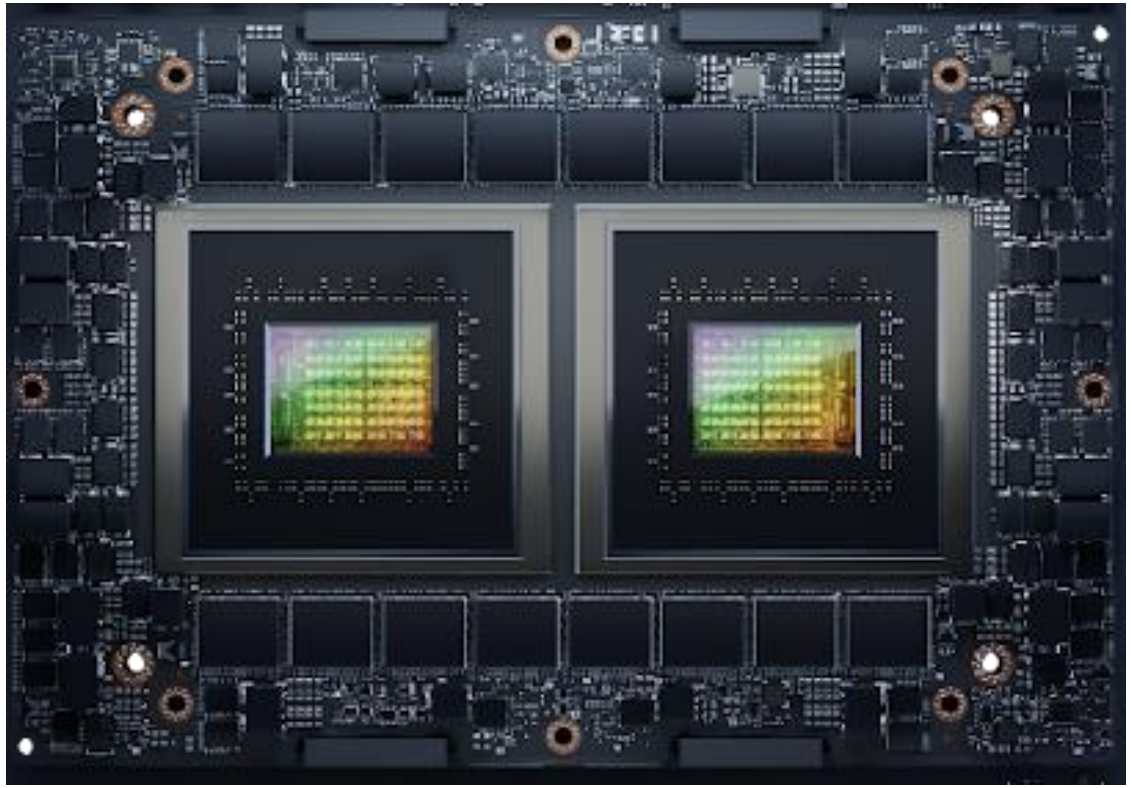
Up to 960GB of data center enhanced LPDDR5X Memory that
delivers up to1TB/s of memory bandwidth

Fast and Flexible CPU IO

Up to 8x PCIe Gen5 x16 interface. PCIe Gen 5 up to 128GB/s
2X more bandwidth compared to PCIe Gen 4

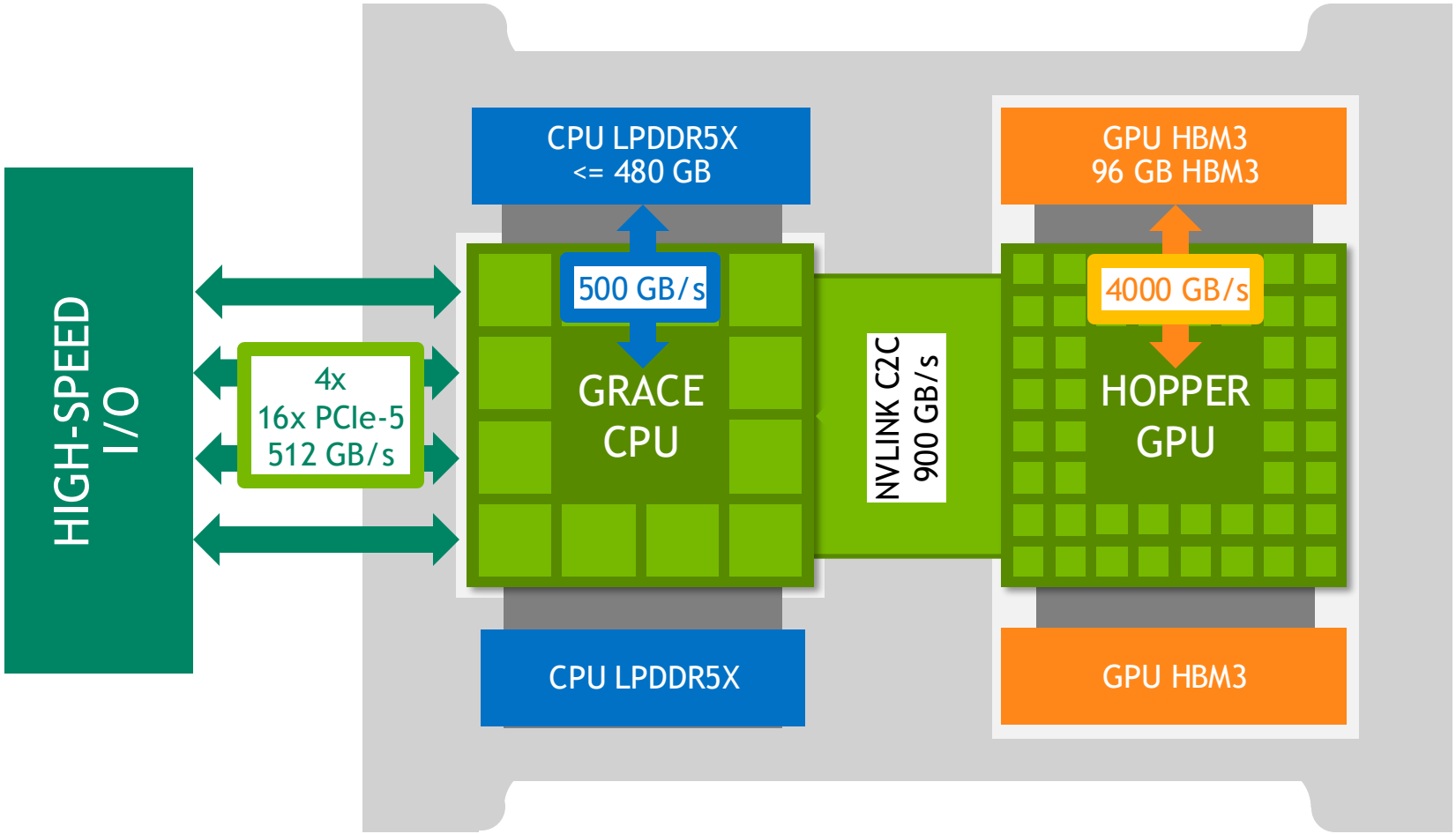
Full NVIDIA Software Stack

AI, Omniverse



Grace Hopper Superchip

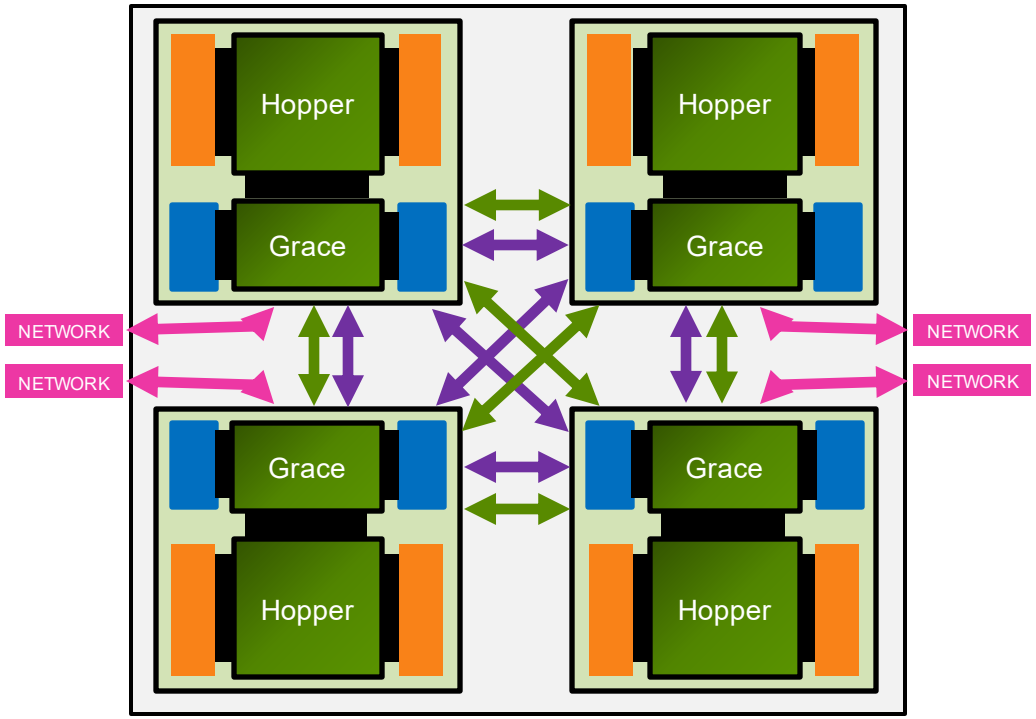
GPU can access CPU memory at CPU memory speeds



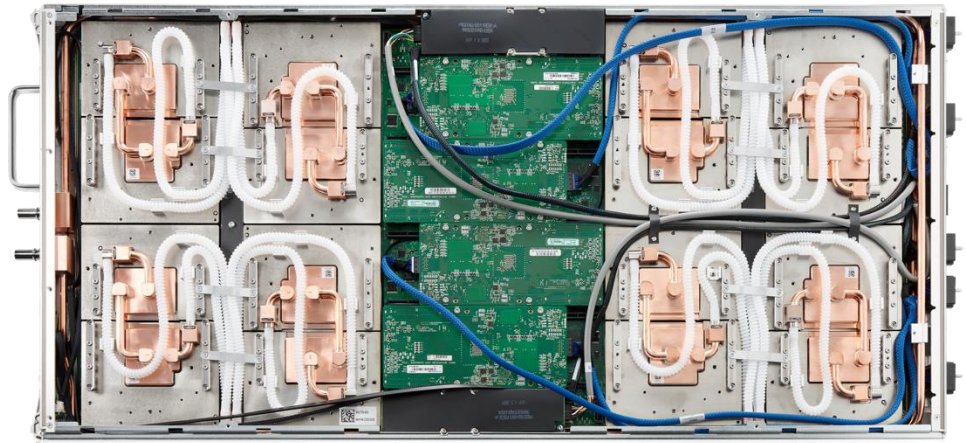
https://resources.nvidia.com/en-us-grace-cpu/nvidia-grace-hopper

NVIDIA Quad GH200

Node architecture for scalable dense supercomputing



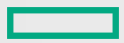
HPE Cray EX254n Accelerator Blade



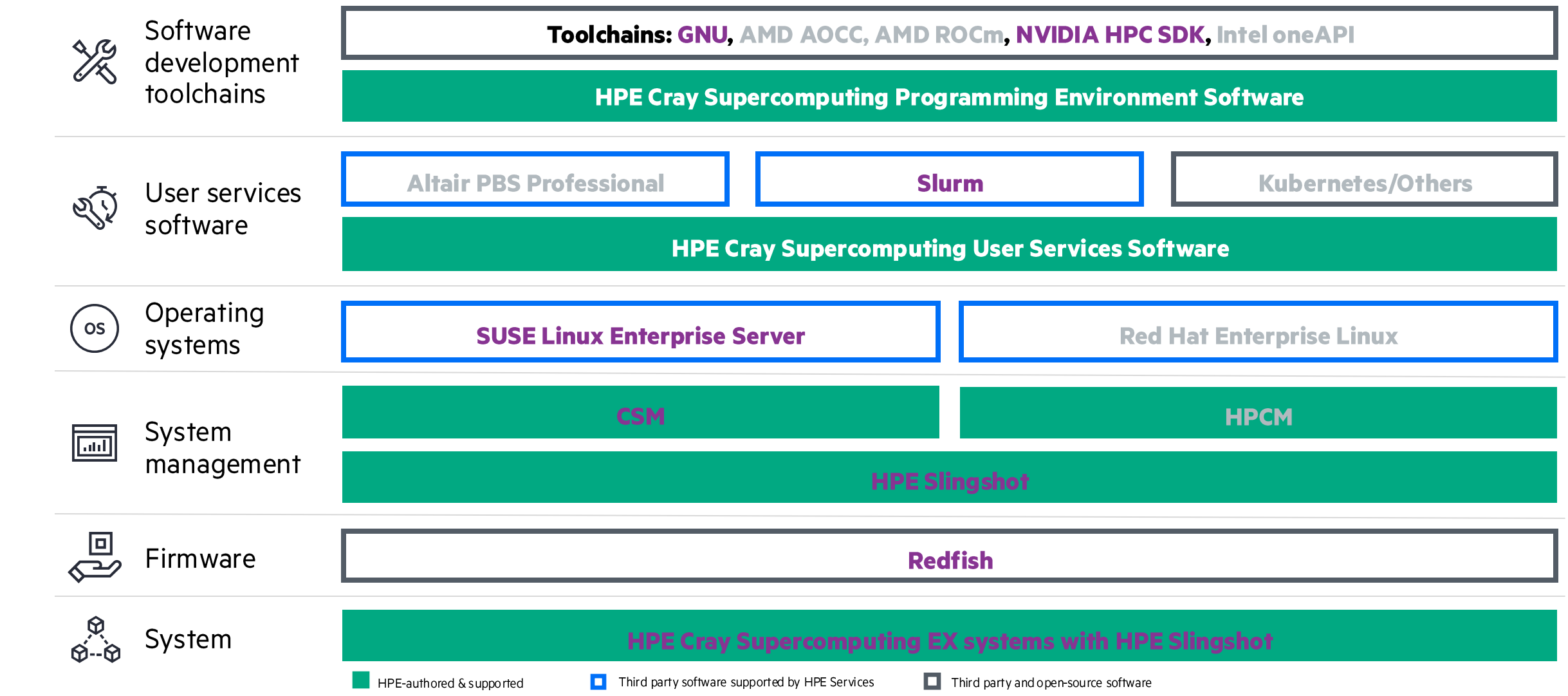
GPU NVLINK
(300 GB)

COHERENT cNVLINK
(200 GB)

Software Environment



HPE Cray EX HPC software portfolio for Isambard AI



Cray Developer Environment for EX

Development

Programming Models

- HPE Cray MPI
- SHMEM
- OpenMP | OpenACC
- AMD ROCm HIP | NVIDIA CUDA
- UPC | Fortran co-arrays
- Chapel

Programming Environments

- Compiling Environment
- GNU
- Intel Programming Environment
- AMD Programming Environment
- NVIDIA HPC SDK

Programming Languages

- C
- C++
- Fortran
- Python
- Chapel
- R

Optimized Libraries

- LibSci (BLAS)
- LAPACK & ScaLAPACK
- LibSci_ACC
- IRT
- FFTW

I/O Libraries

- NetCDF
- HDF5

DL / AI Tools

- Deep Learning Plug-in

Debugging

Comparative Debugger
Compare two versions of an application

GDB for HPC
Parallelized gdb for HPC

Valgrind for HPC
Memory debugging at scale

Stack Trace Analysis Tool
Stack tracing at scale

Tool for Abnormal Termination Processing
Manage core files at scale

- TotalView
- DDT

Performance Analysis & Optimization

Performance Analysis Tool (PAT)
Whole program performance analysis, exposing wide set of indicators, identifying bottlenecks and automatically generating suggestions to improve performance.

Visualization Tool
Complements text reports with summary of performance data in graphs and charts, allowing users to drill down and resolve issues

Code Parallelization Assistant
Reveal hidden potential of an application via code restructuring

Setup & Runtime

Environment Setup

- Tool Enablement (for Spack, CMake, EasyBuild, etc)
- Modules / Lmod

Supported systems:

- HPE Cray EX supercomputers
- HPE Apollo 2000
- HPE Apollo 6500
- HPE ProLiant DL systems
- Previous Cray XC systems

HPE –authored | HPE Added-value to 3rd party | 3rd party

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Components of the Programming Environment

Cray Compiling Environment (CCE)
  Optimizing compilers for Fortran, C, C++ and UPC

Cray Scientific and Math Libraries (CSML)
  High Performance libraries for scientific applications (BLAS, LAPACK, ScaLAPCK, FFTW, NetCDF)

Cray Message Passing Toolkit (CMPT)
  Provides the Message Passing Interface (MPI) and OpenSHMEM

Cray Environment Setup and Compiling Support (CENV)
  Infrastructure to support the development environment.
  Includes compiler drivers, hugepages support and the PE packaging API

Cray Performance Measurement and Analysis Tools (CPMAT)
  Provides tools to analyse performance and behaviour of programs and the PAPI performance API

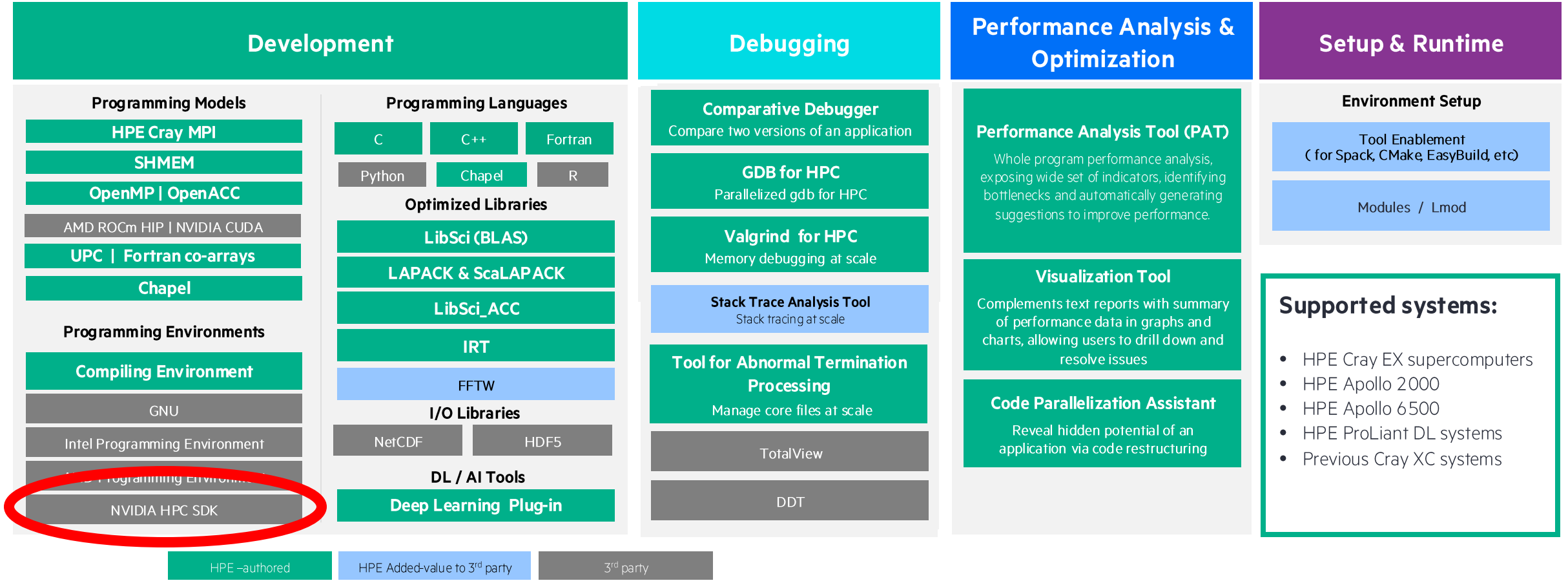
Cray Debugging Support Tools (CDST)
  Provides debugging tools including gdb4hpc and valgrind4hpc

Documentation at https://cpe.ext.hpe.com/docs/24.03/index.html



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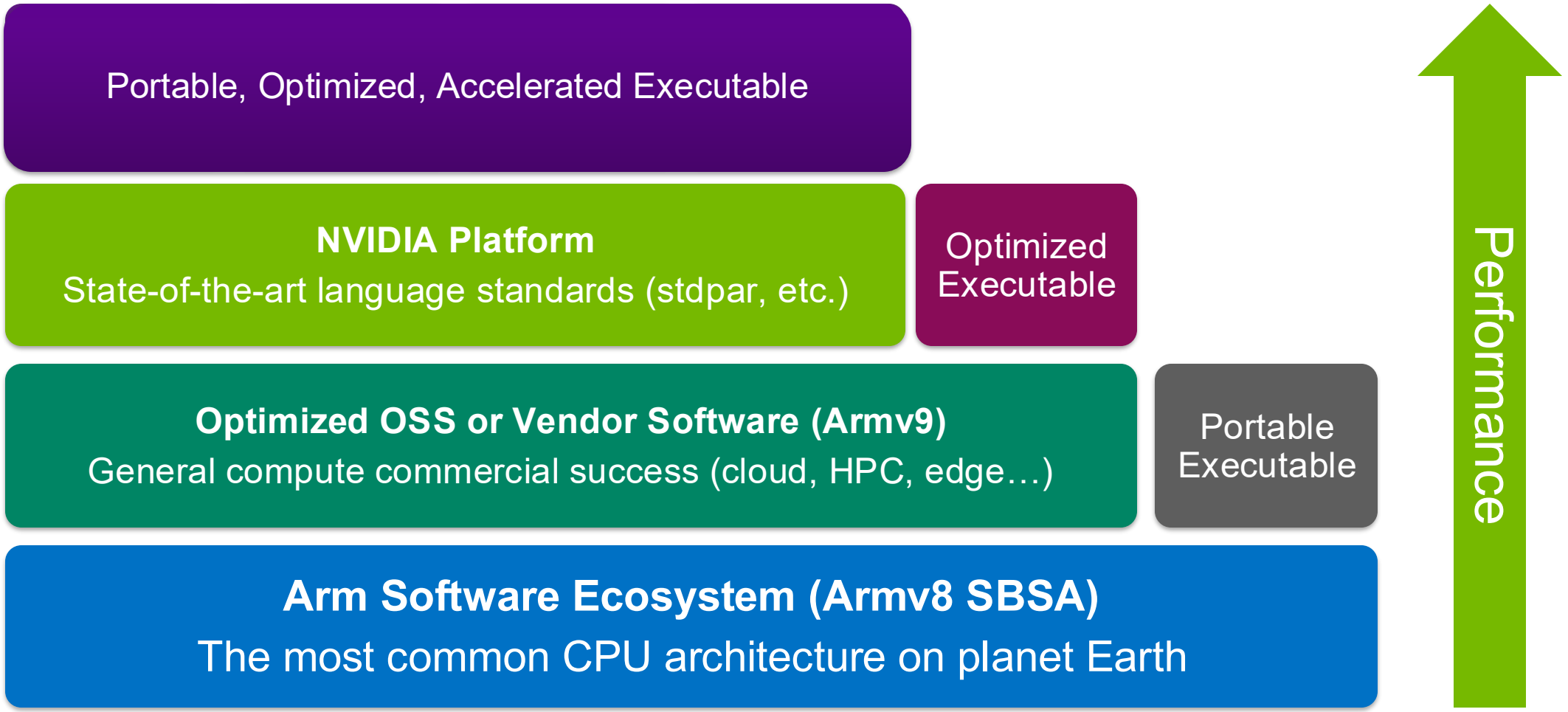
Now lets zoom in to the NVIDIA Software Stack for GH200



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Grace Software Ecosystem is Built on Standards + NVIDIA's Ecosystem

Grace brings the full NVIDIA software stack to Arm.



It Just Works

Most applications will recompile easily and work “out of the box”



Reuse
- NGC
- Your Linux Distros Package Manager

Recompile
- NVIDIA Compiler
- GCC
- LLVM
- Spack or EasyBuild

Run
- High core count
- Threads-per-process
- Update tests

Optimize
- NVIDIA Nsight
- Arm Forge
- Perf, PAPI, TAU, Score-P, ...

Quick Launch

- NGC containerized applications, frameworks, and toolkits
- `./configure && make install`

Compilation Tips

- Most compiler flags are the same:
  - Use `-mcpu=native`
  - Don’t use `-march` or `–mtune`
  - You may need `–fsigned-char`
- Update your unit tests:
  - Aarch64 floating point is as accurate as all other platforms

NVIDIA's Approach to CPU Compilers for Grace

Use the compilers you know and love

- **NVHPC and NVCC**
  - **23.5+ supports Grace and Neoverse V2**
  - Continuously improving performance
- **LLVM Clang**
  - **LLVM16+ supports -mcpu=neoverse-v2**
  - NVIDIA will provide supported builds of Clang as drop-in replacements for mainline Clang
  - NVIDIA contributing to LLVM in open source to maximize Grace application performance
  - Engaging with Arm (Inc.) to improve performance for key CPU benchmarks and applications
- **GCC**
  - **GCC12.2+ supports -mcpu=neoverse-v2**
  - Engaging with Arm (Inc.)
  - Already finding and fixing bugs in mainline GCC

NVHPC

- NVIDIA's innovation space
- Focus on value for workloads

LLVM / Clang

- Required for an excellent CPU software ecosystem
- A foundation for potential compiler innovations

GCC

- Required for an excellent CPU software ecosystem
- Often the default; sets performance expectations

NVIDIA Performance Libraries (NVPL)

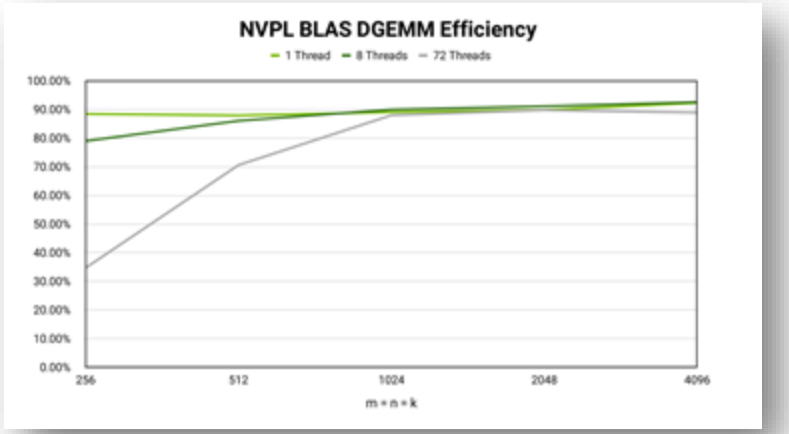
Optimized math libraries for NVIDIA CPUs

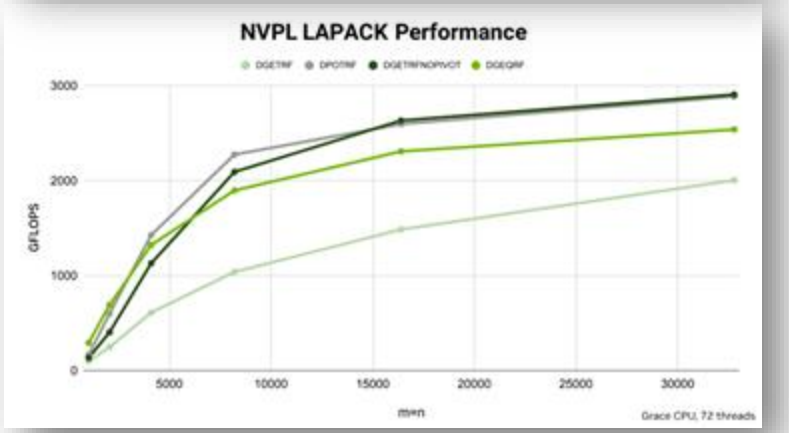
- Easily port applications to NVIDIA's Arm CPUs
- Drop-in replacement for any math library implementing standard interfaces (e.g. Netlib, FFTW)
- New interfaces for high-performance libraries

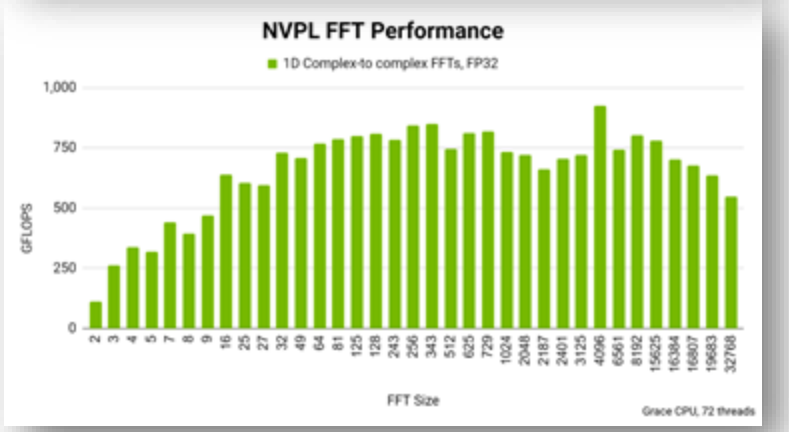
BLAS	LAPACK	PBLAS	SCALAPACK
TENSOR	SPARSE	RAND	FFT

Download Now

www.developer.nvidia.com/nvpl







Porting Applications that use Math Libraries (MKL, OpenBLAS, etc.)

Several library options to choose from

- Prefer Netlib BLAS/LAPACK and FFTW interfaces
  - Building on these interfaces enables compatibility

- **NVPL**

```
gcc -DUSE_CBLAS -ffast-math -mcpu=native -O3 \
    -I/PATH/TO/nvpl/include \
    -L/PATH/TO/nvpl/lib \
    -o mt-dgemm.nvpl mt-dgemm.c \
    -lnvpl_blas_lp64_gomp
```

- **ArmPL**

```
gcc -DUSE_CBLAS -ffast-math -mcpu=native -O3 \
    -I/opt/arm/armpl-23.10.0_Ubuntu-22.04_gcc/include \
    -L/opt/arm/armpl-23.10.0_Ubuntu-22.04_gcc/lib \
    -o mt-dgemm.armpl mt-dgemm.c \
    -larmpl_lp64
```

```
libnvpl_blas_ilp64_gomp.so
libnvpl_blas_ilp64_seq.so
libnvpl_blas_lp64_gomp.so
libnvpl_blas_lp64_seq.so
libnvpl_fftw.so
libnvpl_lapack_ilp64_gomp.so
libnvpl_lapack_ilp64_seq.so
libnvpl_lapack_lp64_gomp.so
libnvpl_lapack_lp64_seq.so
libnvpl_rand_mt.so
libnvpl_rand.so
libnvpl_scalapack_ilp64.so
libnvpl_scalapack_lp64.so
libnvpl_sparse.so
libnvpl_tensor.so
```

- **ATLAS**, **OpenBLAS**, **BLIS**, … Community supported with some optimizations for Neoverse V2.
  - Works on Grace, but unlikely to outperform NVPL and ArmPL. A good compatibility option.

Summary

We have covered:

- A high level overview of the HPE Cray EX4000 Supercomputer Architecture
- The EX245n compute blade with 4-way NVIDIA GH200 Superchip
- The architecture of the GH200
- The software stacks available on this type of system
- In later sessions, you will learn more about how to
  - Write applications using the GH200
  - Write distributed programs to use the Slingshot interconnect
  - Profile applications on GH200



Thank You

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